SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Student Attendance Management System

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1.0	05-09-2026	Instructor	Initial complete SRS adapted from the ATM model answer	Pending

Approvals

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1. Introduction

1.1 Purpose

This document specifies the functional, non-functional, interface, security, and verification requirements for a Student Attendance Management System (SAMS). The system is intended to digitize attendance recording, reduce manual calculation, provide accurate attendance percentages, and automatically generate attendance reports for academic use.

1.2 Scope

SAMS will support administrators, faculty members, and students. It will manage users, courses, class rosters, attendance sessions, attendance records, corrections, attendance percentages, low-attendance identification, and report generation/export. It may be implemented using Python (Django/Flask) or Java (Spring Boot) with a relational database. The SRS is implementation-independent unless a technology is explicitly stated as a constraint.

Out of scope: biometric hardware design, payroll/employee attendance, university ERP internals beyond exposed integration APIs, and automatic grading based on attendance.

1.3 Audience

Intended readers are developers, QA engineers, database designers, system administrators, instructors, students, project evaluators, and system integrators.

1.4 Definitions and Acronyms

Term	Meaning
SAMS	Student Attendance Management System
Admin	Authorized administrator who manages master data and access
Faculty	Instructor authorized to create sessions and record attendance
Student	Learner enrolled in a course who can view own attendance
Attendance Session	A specific class meeting for which attendance is recorded
FR	Functional Requirement
NFR	Non-functional Requirement
RBAC	Role-Based Access Control
API	Application Programming Interface
RTM	Requirements Traceability Matrix

2. Overall Description

2.1 Product Perspective

SAMS is a web-based information system. Users access it through a browser. The application server handles authentication, attendance logic, reporting, and APIs; a database stores users, courses, enrollments, sessions, attendance records, and audit events. Optional services can provide email/SMS notifications and institutional ERP integration.

2.2 Major Product Functions

- Authenticate users and enforce role-based permissions.
- Manage student, faculty, course, section, and enrollment data.
- Create scheduled or ad-hoc attendance sessions.
- Record Present, Absent, Late, or Excused status for enrolled students.
- Allow controlled attendance corrections with reason and audit history.
- Calculate course-wise attendance counts and percentages automatically.
- Identify students below a configurable attendance threshold.
- Generate class, student, course, date-range, and shortage reports.
- Export reports in PDF/CSV format.
- Provide dashboards and attendance history to authorized users.
- Maintain audit logs and operational logs.
- Optionally send low-attendance notifications and integrate with an institutional API.

2.3 User Roles and Characteristics

Role	Characteristics / Responsibilities
Student	Basic web usage; can view own attendance, course-wise percentage, and notifications.
Faculty	Creates sessions, marks/corrects attendance for assigned courses, and generates reports.
Administrator	Manages users, courses, enrollments, thresholds, permissions, reports, and audit records.
System/Integration Service	Optional service account for scheduled jobs, notifications, backups, and ERP/API synchronization.

2.4 Operating Environment

- Server: Linux/Windows server capable of running Django/Flask or Spring Boot.
- Client: modern desktop/mobile browser with JavaScript enabled.
- Database: relational DBMS such as PostgreSQL/MySQL (or equivalent).
- Network: campus LAN/WAN/Internet over HTTPS.
- Deployment may be on-premises or cloud-hosted.

2.5 Constraints and Assumptions

- Only authorized users may modify attendance data.
- Each attendance record must be linked to a valid student, course/section, and session.
- A student must be enrolled in a section before attendance can be recorded for that section.
- Attendance threshold (for example 75%) shall be configurable; no fixed percentage is assumed by the SRS.
- The implementation shall use a supported framework such as Django/Flask or Spring Boot, as required by the project allocation.
- Server time is treated as the authoritative timestamp for attendance and audit records.

3. External Interface Requirements

3.1 User Interfaces

- Login page with username/email and password.
- Role-specific dashboard showing relevant courses, sessions, alerts, and attendance summaries.
- Faculty attendance screen with roster, date/time, session details, and status controls.
- Admin screens for user/course/section/enrollment management.
- Student view showing own subject-wise attendance and attendance history.
- Reports screen with filters, preview, and PDF/CSV export.
- Clear validation/error messages and confirmation prompts for destructive changes.
- Responsive design and keyboard-accessible controls.

3.2 Hardware Interfaces

No mandatory special-purpose hardware is required. The minimum client interface is a computer or mobile device with a modern browser. If QR, RFID, biometric, or smart-card attendance is later added, it shall be treated as an extension and integrated through a defined device/API interface.

3.3 Software Interfaces

- Relational Database interface for persistent storage and transactional updates.
- Optional institutional/ERP API for importing student, faculty, course, and enrollment data.
- Optional email/SMS gateway for attendance warnings and report notifications.
- PDF/CSV generation library for downloadable reports.
- Authentication/session management component supplied by the chosen framework.

3.4 Communications

- All browser-to-server traffic shall use HTTPS in production.
- REST/JSON may be used for client/API communication.
- API calls shall use authenticated requests and standard HTTP status codes.
- Network errors shall produce safe user messages without exposing stack traces or sensitive server details.

Project minimums from the provided SRS template are satisfied below: at least 15 functional requirements, at least 5 non-functional requirements, at least 2 security objectives, and at least 5 security requirements.

4. System Features (Detailed)

Each functional requirement uses an ID of the form SAMS-F-###. Acceptance criteria are deliberately testable so that every requirement can be linked to a verification test in the RTM.

4.1 Authentication and Access Control

Secure login and role-based access for students, faculty, and administrators. Requirement IDs: SAMS-F-001 to SAMS-F-002.

4.2 Master Data and Enrollment Management

Administration of users, courses, sections, faculty assignments, and enrollments. Requirement IDs: SAMS-F-003 to SAMS-F-005.

4.3 Attendance Session and Recording

Creation of class sessions, roster display, attendance marking, duplicate prevention, and controlled corrections. Requirement IDs: SAMS-F-006 to SAMS-F-010.

4.4 Attendance Calculation and Viewing

Automatic calculation and authorized viewing of attendance summaries and details. Requirement IDs: SAMS-F-011 to SAMS-F-013.

4.5 Reports, Shortage Identification, Notifications and Audit

Threshold-based shortage detection, report generation/export, optional notifications, and audit logging. Requirement IDs: SAMS-F-014 to SAMS-F-018.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref
SAMS-F-001	Authenticate a user using valid credentials and create a session associated with the user role.	Functional	High	All users	Valid credentials open the correct role dashboard; invalid credentials are rejected. TC-AUTH-01
SAMS-F-002	Enforce RBAC so that Students, Faculty, and Administrators can access only permitted functions and data.	Functional	High	Security/Admin	Unauthorized URL/API access returns access denied and no protected data. TC-AUTH-02
SAMS-F-003	Allow an Administrator to create, view, update, activate, and deactivate student and faculty accounts.	Functional	High	Admin	Account changes persist and deactivated users cannot log in. TC-ADM-01
SAMS-F-004	Allow an Administrator to create and manage courses, sections, semesters, and faculty assignments.	Functional	High	Admin	Valid master data can be created/edited and invalid duplicate mappings are rejected. TC-ADM-02
SAMS-F-005	Allow an Administrator to enroll and remove students from course sections, individually or by supported bulk import.	Functional	High	Admin	Roster reflects successful enrollment changes and invalid IDs are reported. TC-ADM-03
SAMS-F-006	Allow authorized Faculty to create an attendance session for an assigned section with date, start time, and optional topic/period.	Functional	High	Faculty	Session is created only for a faculty-assigned section and appears in session history. TC-ATT-01
SAMS-F-007	Display the current enrolled roster for the selected attendance session.	Functional	High	Faculty	Roster contains all active enrolled students and excludes non-enrolled students. TC-ATT-02
SAMS-F-008	Allow Faculty to mark each rostered student as Present, Absent, Late, or Excused and save the attendance.	Functional	High	Faculty	Saved statuses are retrievable and one status exists per student/session. TC-ATT-03
SAMS-F-009	Prevent duplicate attendance records for the same student and attendance session.	Functional	High	Data integrity	Second duplicate insert is blocked or converted into an authorized update. TC-ATT-04
SAMS-F-010	Allow authorized Faculty/Admin to correct a saved attendance record by supplying a reason; preserve who changed it and when.	Functional	High	Faculty/Admin	Correction is visible in current record and audit history contains old/new values, actor, time, reason. TC-ATT-05
SAMS-F-011	Automatically calculate attended classes, total counted classes, and attendance percentage per student per course/section.	Functional	High	Students/Faculty	Calculated values match test data and update after attendance changes. TC-CALC-01
SAMS-F-012	Allow Students to view only their own course-wise attendance percentage and session-level history.	Functional	High	Student	Student can view own data but cannot retrieve another student's details. TC-STU-01
SAMS-F-013	Allow Faculty and Admin to view class-wise attendance summaries and individual student attendance details for authorized sections.	Functional	Medium	Faculty/Admin	Summary totals and drill-down data match stored attendance. TC-RPT-01
SAMS-F-014	Identify students whose attendance is below a configurable threshold for a selected course/section.	Functional	High	Faculty/Admin	Shortage list matches threshold calculation on known dataset. TC-RPT-02
SAMS-F-015	Generate attendance reports filtered by course/section, student, attendance status, and date range as permitted by role.	Functional	High	Faculty/Admin	Report results exactly match selected filters. TC-RPT-03
SAMS-F-016	Export an authorized attendance report in CSV and PDF format with generation timestamp and applied filters.	Functional	Medium	Faculty/Admin	Downloaded files open successfully and contain the same rows/summary as the on-screen report. TC-RPT-04
SAMS-F-017	Send or queue a low-attendance notification when enabled and when a student crosses the configured threshold.	Functional	Medium	Student/Admin	A test crossing creates one notification event containing course and current percentage. TC-NOTIF-01
SAMS-F-018	Record audit events for login failures, attendance creation/correction, user management, threshold changes, and report exports.	Functional	High	Security/Admin	Audit search shows timestamp, actor, action, target, and outcome for tested events. TC-AUD-01

5. Non-functional Requirements (Detailed)

Non-functional requirements define measurable quality constraints rather than user-visible business functions.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
SAMS-NF-001	For normal academic load, 95% of interactive page/API requests shall complete within 3 seconds, excluding large report exports.	Performance	High	Load test shows 95th percentile <= 3 s. TC-PERF-01
SAMS-NF-002	The production service shall achieve at least 99.5% monthly availability, excluding approved maintenance windows.	Reliability	High	Monitoring report shows >= 99.5% monthly uptime. TC-REL-01
SAMS-NF-003	The system shall support at least 500 concurrent authenticated users and 10,000 student records without functional failure in the reference deployment.	Scalability	Medium	Capacity test completes with no data loss and response target remains acceptable. TC-PERF-02
SAMS-NF-004	Core user workflows shall be usable on current major desktop and mobile browsers and support keyboard navigation and readable form labels.	Usability/Accessibility	Medium	Cross-browser and accessibility checklist passes. TC-UX-01
SAMS-NF-005	Attendance and audit data shall be backed up at least daily; restoration of a verified backup shall be demonstrated in a test environment.	Maintainability/Recovery	High	Scheduled backup exists and restore test recovers the expected dataset. TC-OPS-01
SAMS-NF-006	The application shall validate inputs and maintain database integrity so that committed attendance operations are atomic and internally consistent.	Data Integrity	High	Fault/validation tests do not create partial or orphaned attendance records. TC-DATA-01

5.1 Security

5.1.1 Security Objectives

SO-1 — Confidentiality and least privilege: Protect credentials and attendance information from unauthorized disclosure, and ensure users can access only data required by their role.

SO-2 — Integrity and accountability: Prevent unauthorized or untraceable changes to attendance records, and maintain sufficient audit evidence to identify who performed sensitive actions and when.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
SAMS-SR-001	All production client-server and API communication shall use HTTPS with TLS 1.2 or higher.	Security	High	HTTP is redirected/blocked; security scan confirms TLS 1.2+ only. TC-SEC-01
SAMS-SR-002	Passwords shall never be stored in plaintext and shall be stored using the framework's approved adaptive salted password hashing mechanism.	Security	High	Database inspection shows only salted password hashes; plaintext password search returns none. TC-SEC-02
SAMS-SR-003	The system shall enforce server-side authorization for every protected page and API endpoint using the authenticated role and object-level permissions.	Security	High	Student/faculty attempts to access unauthorized objects/endpoints are denied. TC-SEC-03
SAMS-SR-004	The system shall protect authenticated sessions using Secure, HttpOnly cookies (where cookies are used), session expiration, logout invalidation, and CSRF protection for state-changing browser requests.	Security	High	Cookie/security inspection and CSRF/session tests pass. TC-SEC-04
SAMS-SR-005	Sensitive administrative and attendance modifications shall create tamper-resistant audit entries containing actor, timestamp, action, target, outcome, and relevant change metadata without storing passwords.	Security	High	Audit test verifies all required fields and confirms passwords are absent. TC-SEC-05
SAMS-SR-006	Repeated failed login attempts shall be rate-limited or temporarily throttled/locked according to a configurable security policy.	Security	Medium	Automated repeated-failure test triggers throttling/lock behavior. TC-SEC-06

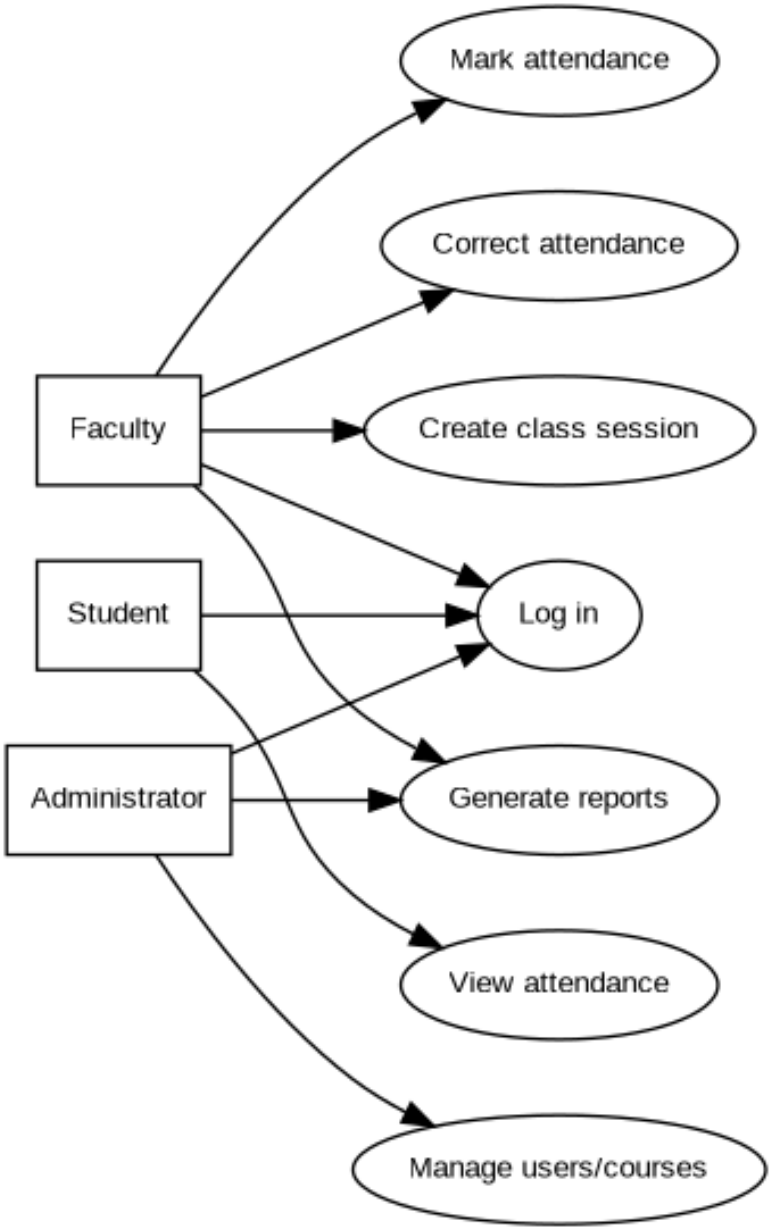
6. Quality Attributes & Acceptance Tests

Acceptance exit criteria: all High-priority functional and security requirements shall be implemented and pass their referenced tests; no open Critical defects shall remain; required NFR measurements shall pass; and the RTM shall show a test reference for every listed requirement.

- Authentication & RBAC suite: TC-AUTH-01..02
- Administration/master-data suite: TC-ADM-01..03
- Attendance suite: TC-ATT-01..05
- Calculation/student view suite: TC-CALC-01, TC-STU-01
- Reporting/notification/audit suite: TC-RPT-01..04, TC-NOTIF-01, TC-AUD-01
- Performance/reliability/recovery/data suite: TC-PERF-01..02, TC-REL-01, TC-OPS-01, TC-DATA-01
- Security suite: TC-SEC-01..06
- Usability/accessibility suite: TC-UX-01

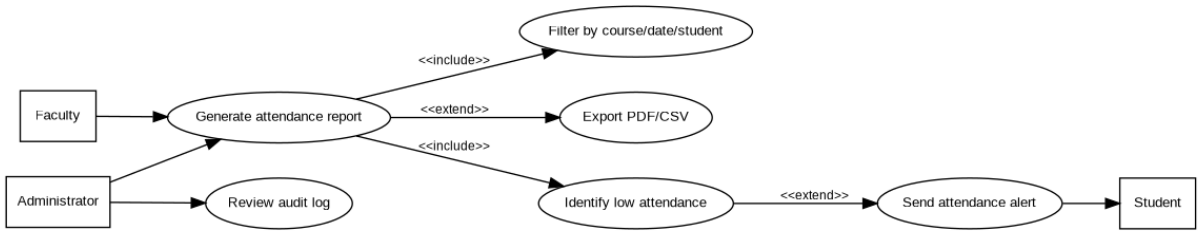
7. System Models and Diagrams

7.1 UML Use-Case Diagram 1 — Core Attendance Operations



This diagram shows the primary actors and the core interactions for authentication, attendance recording, attendance viewing, administration, and reporting.

7.2 UML Use-Case Diagram 2 — Reporting, Shortage and Audit



This diagram focuses on report generation, filtering/export, shortage identification, optional notification, and administrative audit review.

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
SAMS-F-001	Authenticate a user using valid credentials a...	4 / DS-AuthModule	AuthModule	TC-AUTH-01	N	Not implemented / to be verified
SAMS-F-002	Enforce RBAC so that Students, Faculty, and A...	4 / DS-Auth/RBAC	Auth/RBAC	TC-AUTH-02	N	Not implemented / to be verified
SAMS-F-003	Allow an Administrator to create, view, updat...	4 / DS-UserAdmin	UserAdmin	TC-ADM-01	N	Not implemented / to be verified
SAMS-F-004	Allow an Administrator to create and manage c...	4 / DS-CourseAdmin	CourseAdmin	TC-ADM-02	N	Not implemented / to be verified
SAMS-F-005	Allow an Administrator to enroll and remove s...	4 / DS-Enrollment	Enrollment	TC-ADM-03	N	Not implemented / to be verified
SAMS-F-006	Allow authorized Faculty to create an attenda...	4 / DS-Session	Session	TC-ATT-01	N	Not implemented / to be verified
SAMS-F-007	Display the current enrolled roster for the s...	4 / DS-Roster	Roster	TC-ATT-02	N	Not implemented / to be verified
SAMS-F-008	Allow Faculty to mark each rostered student a...	4 / DS-Attendance	Attendance	TC-ATT-03	N	Not implemented / to be verified
SAMS-F-009	Prevent duplicate attendance records for the ...	4 / DS-Attendance	Attendance	TC-ATT-04	N	Not implemented / to be verified
SAMS-F-010	Allow authorized Faculty/Admin to correct a s...	4 / DS-AttendanceAudit	AttendanceAudit	TC-ATT-05	N	Not implemented / to be verified
SAMS-F-011	Automatically calculate attended classes, tot...	4 / DS-AttendanceCalc	AttendanceCalc	TC-CALC-01	N	Not implemented / to be verified
SAMS-F-012	Allow Students to view only their own course-...	4 / DS-StudentPortal	StudentPortal	TC-STU-01	N	Not implemented / to be verified
SAMS-F-013	Allow Faculty and Admin to view class-wise at...	4 / DS-Reports	Reports	TC-RPT-01	N	Not implemented / to be verified
SAMS-F-014	Identify students whose attendance is below a...	4 / DS-Reports	Reports	TC-RPT-02	N	Not implemented / to be verified
SAMS-F-015	Generate attendance reports filtered by cours...	4 / DS-Reports	Reports	TC-RPT-03	N	Not implemented / to be verified
SAMS-F-016	Export an authorized attendance report in CSV...	4 / DS-Export	Export	TC-RPT-04	N	Not implemented / to be verified
SAMS-F-017	Send or queue a low-attendance notification w...	4 / DS-Notification	Notification	TC-NOTIF-01	N	Not implemented / to be verified
SAMS-F-018	Record audit events for login failures, atten...	4 / DS-Audit	Audit	TC-AUD-01	N	Not implemented / to be verified
SAMS-NF-001	For normal academic load, 95% of interactive ...	5 / DS-Performance	Performance	TC-PERF-01	N	Pending verification
SAMS-NF-002	The production service shall achieve at least...	5 / DS-Reliability	Reliability	TC-REL-01	N	Pending verification
SAMS-NF-003	The system shall support at least 500 concurr...	5 / DS-Scalability	Scalability	TC-PERF-02	N	Pending verification
SAMS-NF-004	Core user workflows shall be usable on curren...	5 / DS-Usability-Accessibility	Usability/Accessibility	TC-UX-01	N	Pending verification
SAMS-NF-005	Attendance and audit data shall be backed up ...	5 / DS-Maintainability-Recovery	Maintainability/Recovery	TC-OPS-01	N	Pending verification
SAMS-NF-006	The application shall validate inputs and mai...	5 / DS-Data Integrity	Data Integrity	TC-DATA-01	N	Pending verification
SAMS-SR-001	All production client-server and API communic...	5.1 / DS-Security	Security	TC-SEC-01	N	Pending verification
SAMS-SR-002	Passwords shall never be stored in plaintext ...	5.1 / DS-Security	Security	TC-SEC-02	N	Pending verification
SAMS-SR-003	The system shall enforce server-side authoriz...	5.1 / DS-Security	Security	TC-SEC-03	N	Pending verification
SAMS-SR-004	The system shall protect authenticated sessio...	5.1 / DS-Security	Security	TC-SEC-04	N	Pending verification
SAMS-SR-005	Sensitive administrative and attendance modif...	5.1 / DS-Security	Security	TC-SEC-05	N	Pending verification
SAMS-SR-006	Repeated failed login attempts shall be rate-...	5.1 / DS-Security	Security	TC-SEC-06	N	Pending verification

RTM status legend: N = Not yet verified, P = Passed, A = Accepted/Approved. Update these values during implementation and testing.

Appendix A — How to Read the Requirements

Functional Requirements (SAMS-F-###) describe what the system must do. Example: create a class session, mark attendance, calculate percentage, or generate a report.

Non-functional Requirements (SAMS-NF-###) describe measurable quality properties of the system, such as performance, reliability, scalability, accessibility, recovery, and data integrity.

Security Objectives are high-level protection goals. Security Requirements (SAMS-SR-###) convert those goals into testable technical requirements such as HTTPS, password hashing, authorization, session security, audit logging, and login throttling.

Acceptance Criteria state how a requirement will be considered satisfied. Test Case references provide traceability from the requirement to verification. The RTM links each requirement to a design/module and its test case so that missing implementation or testing can be detected.